# OPTICAL LOSS BUDGET STANDARD

Standard OSP-207 | Revision D | Effective 2 March 2026 | Owner: Outside Plant Engineering | Supersedes Revision C

## 1. Scope

This standard sets the component loss values and the received-power acceptance thresholds used to verify Gigabit Passive Optical Network (GPON) distribution plant. It applies at the 1490 nm downstream wavelength. All fibre in the access plant is ITU-T G.652.D single mode. Revision D restates the mechanical splice allowance, which was previously quoted as a range.

## 2. Component loss values

These are design values. A component whose measured loss exceeds its design value is out of specification and shall be recorded as such.

| Component | Design loss | Unit |
|---|---|---|
| Single-mode fibre attenuation at 1490 nm | 0.25 | dB per km |
| Fusion splice | 0.10 | dB each |
| Mechanical splice | 0.30 | dB each |
| Mated connector pair | 0.50 | dB each |
| 1:8 optical splitter, insertion loss | 10.50 | dB |
| 1:4 optical splitter, insertion loss | 7.20 | dB |

## 3. Transmit reference

Optical line terminal (OLT) transmit power at the passive optical network (PON) port is commissioned and held at **+2.50 dBm**. End-to-end path loss is the difference between that transmit power and the power received at the optical network terminal (ONT).

## 4. Received-power acceptance thresholds

Every in-service ONT is classified from its measured received power as set out below. Boundaries are inclusive as written.

| Classification | Measured ONT received power |
|---|---|
| PASS | greater than or equal to -25.00 dBm |
| MARGINAL | below -25.00 dBm and greater than or equal to -27.00 dBm |
| FAIL | below -27.00 dBm |

## 5. Measurement tolerance

Power meters and optical time-domain reflectometers (OTDRs) in field service carry a combined uncertainty of plus or minus 1.00 dB. A difference between measured loss and design loss of 1.00 dB or less is therefore within tolerance and shall not be raised as a fault. A difference greater than 1.00 dB indicates excess loss that is physically present in the plant and shall be investigated.

## 6. OTDR test method for distribution legs

A distribution leg is tested by shooting an OTDR from the fibre distribution hub (FDH) output port toward the multiport service terminal (MST). The trace datum, 0.00 km, is the FDH output port. Because the MST houses a splitter, the trace terminates at the MST input connector: **it cannot see any fibre, connector or splice downstream of the MST, and that includes all subscriber drop cable.** Excess loss downstream of an MST is therefore invisible to this test and must be isolated from received-power measurements taken at the subscriber terminal.

## 7. Recording

An event recorded by an OTDR is reported by its distance from the trace datum. The trace does not identify whether a splice is fusion or mechanical; that is established from the plant record for the leg concerned, and the design value appropriate to the splice type is the one that applies.